

1(a)	<i>any three from:</i> <i>secondary structure</i> I α-helix 1 (many) <u>β-pleated</u> / <u>beta-pleated</u>, sheets ; R 'B' 2 random structure / irregular structures / loops / beta turns / AW ; <i>tertiary structure</i> 3 folding / coiling, of, (each) polypeptide chain(s) / secondary structure ; R <i>idea of</i> polypeptide chains interacting (quaternary structure) 4 <i>ref. to</i> globular A description, e.g. spherical I circular / round or <i>ref. to</i> 3D, shape / structure ; A 3D arrangement	3
1(b)(i)	X – site of synthesis of, (light and heavy) polypeptides ; A protein(s) A transport / modification, of, polypeptides / proteins A assembly of polypeptides / translation R answers that name the Golgi body Y – production of ATP ; R 'produce / create / AW, energy' A release of energy / provide energy	2
1(b)(ii)	cell / membrane, fusion / AW ; I 'mix' (named) fusogen / hybridogen used ; e.g. polyethylene glycol / electrofusion / electric current A PEG for polyethylene glycol (between) plasma cell / (activated) B-lymphocyte / (activated) B-cell / splenocyte, and, tumour / cancer / myeloma, cell ; R β cells	3

Question	Answer	Marks
1(b)(iii)	<i>any three from:</i> 1 some mAbs act directly on target cells / some mAbs work indirectly to kill cells / mAbs do not damage other (non-target) cells ; 2 by binding to, <u>specific</u> / <u>complementary</u>, antigens/cell surface receptors ; 3 (named), drugs / radioactive isotopes, can be attached to mAbs ; A 'tagged' I labelled 4 enzymes can be attached to mAbs ; 5 so drug can be activated at site of action (linked to mp4) ; 6 bispecific mAbs attach two cells together ; 7 <i>ref.</i> to interrupting cell signalling ; 8 use of mAbs for passive immunity ; A described <i>in context of therapeutic antibody for treatment of disease</i> 9 stimulating / AW, immune system / phagocytes / macrophages / T-lymphocytes, to kill, cancer cells ; 10 name of a cancer or autoimmune disease that is treated with mAbs ;	3